

Compiler infrastructure engineer with professional LLVM 22 and C#.NET static-analysis experience. Public compiler/runtime projects date to 2022, with CIL and native x86-64 code generation by 2023 and an AST → IR → x64 pipeline by 2024. I now build UniversalToolchain around deterministic compiler composition, with 1,324/1,324 tests passing.

2022: Compilers / VMs · 2023: CIL + native x86-64 · 2024–2025: AST / IR / backends → toolchain · 2026 → : Professional LLVM + static analysis

| PROFESSIONAL EXPERIENCE                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                     | SKILLS                                                                                                                                                                                  |
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| 2026 — present                                                                                                                                                                                                                                                                                                                                                                                                      | <b>ISP RAS — Static Analysis Engineer</b> <ul style="list-style-type: none"><li>Contribute to SharpChecker, ISP RAS's C#.NET static-analysis platform.</li></ul>                                                                                                                                                                                                    | <b>Compiler Infrastructure</b><br>dependency/conflict resolution, component/provider selection, deterministic pass ordering, typed artifact routing, pre-execution checks               |
| July — August 2026                                                                                                                                                                                                                                                                                                                                                                                                  | <b>MCST — Compiler Engineering Intern</b> <ul style="list-style-type: none"><li>Implemented a LICM pass for LLVM 22 in C++; hoisting conditions check loop structure, memory access, side effects, and speculative safety.</li><li>Built a Python harness comparing program behavior before/after optimization and checking transform/no-transform cases.</li></ul> | <b>Compiler / IR</b><br>C++23, LLVM 22/IR, CFG/SSA, loop and interprocedural analysis, transformation applicability                                                                     |
| <b>Additional product experience</b> — Diploma.school — client development: jQuery dropdown interactions, responsive mobile/desktop navigation, and UI regression fixes; confirmed no later than January 2025.                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                     | <b>Backends / Runtime</b><br>.NET, CIL/MSIL, native x86-64, SysV ABI, stack frames/calling conventions, register allocation, interpreter/runtime composition                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                     | <b>Verification</b><br>regression suites, LLVM Verifier, interpreter/backend parity, differential execution                                                                             |
| PROJECTS                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                     | EDUCATION                                                                                                                                                                               |
| <b>UniversalToolchain</b><br>Language features resolve dependencies and conflicts, select capability/runtime providers, obtain deterministic contribution ordering, and build typed artifact routes before runtime materialization.<br>Current manifest: 1,324/1,324 tests; dedicated checks cover interpreter/CIL parity and configuration integrity before execution.                                             |                                                                                                                                                                                                                                                                                                                                                                     | HSE University — Software Engineering, 2026–2030                                                                                                                                        |
| <b>LLVM Interprocedural Global IV Optimization</b><br>An LLVM 22 pass localizes supported induction-like globals after interprocedural effect analysis, affine evolution, and a separate applicability check; APInt models fixed-width wraparound.<br>29 positive/negative regression cases cover LLVM Verifier checks, structural expectations, idempotence, and observable before/after behavior.                 |                                                                                                                                                                                                                                                                                                                                                                     | <b>RECOGNITION</b>                                                                                                                                                                      |
| <b>x86-64 Compiler Backend &amp; Register Allocation</b><br>SSA/CFG validation → liveness/interference → register allocation/spills → phi lowering → SysV x86-64 emission.<br>An independent assignment verifier plus native-vs-interpreter differential execution; tests cover branches, loops, spills, and parallel phi moves; a separate public native-codegen lineage starts with Visk (2023) and Wist4 (2024). |                                                                                                                                                                                                                                                                                                                                                                     | <b>LangDev'26 — Build the Language, Then Make the Abstractions Disappear</b><br>Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga. |
|                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                     | <b>UniversalToolchain — Baltic Science and Engineering Competition</b><br>First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain.                             |
|                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                     | <b>MEPhI Junior</b><br>Overall winner; First Degree Diploma, 96/100 for UniversalToolchain.                                                                                             |
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